4780/6780: Fundamentals of Data Science

Kiril Kuzmin

Spring, 2025

E-mail: `kkuzmin1@gsu.edu`

Office Hours: M 2:00–4:00pm
Office: 25 Park Place 750

Class Hours: M,W 5:30–7:15pm
Class Room: Langdale Hall 215

Course Description

This course provides students with essential concepts, principles, and tools to extract and generalize knowledge from data. Students will develop a comprehensive skill set, including data processing, statistics, and machine learning, and gain a solid understanding of how these skills integrate to solve real-world problems. The course systematically introduces key topics in data science, including:

1. Principles of data processing and representation;
2. Theoretical foundations and recent advancements in data science;
3. Modeling and algorithms;
4. Evaluation techniques.

The focus will be on the breadth of these topics rather than on in-depth exploration. Real-world engineering problems and datasets will be used to illustrate the advantages and limitations of various algorithms, allowing students to compare their effectiveness and efficiency and to discern the contexts in which specific algorithms are most appropriate.

Subgroups

Our class is composed of three subgroups that come together as a one large group:

CRN	Section
19296	CSC 4780
19297	CSC 6780
19298	DSCI 4780

Tentative Weekly Schedule

Week 1: Introduction to Data Science

- Overview of data science

- Types of learning
- Predictive data analytics
- Introduction to `numpy` for numerical computations

Week 2: Data Manipulation and Analysis

- Introduction to `pandas` for data manipulation and analysis
- Problem-solving with `numpy` and `pandas`

Week 3: Data Exploration and Visualization

- Data visualization using `matplotlib` and `seaborn`
- Exploratory data analysis

Week 4: Data Preparation and Presentation

- Data preparation techniques
- Principles of effective data presentation and visualization

Week 5: Information-Based Learning

- Entropy, information gain, and Gini index
- Decision trees and the ID3 algorithm
- Successors of ID3: C4.5 and CART (Classification and Regression Trees)

Week 6: Similarity-Based Learning

- Introduction to k-NN (k-Nearest Neighbors)
- Midterm review

Week 7: Probability-Based Learning and Midterm

- Naive Bayes model
- Midterm exam

Week 8: Error-Based Learning

- Linear and logistic regression
- Introduction to SVM (Support Vector Machine) and kernels

Week 9: Feature Selection and Engineering

- Correlation analysis and mutual information

- Encoding techniques and feature creation/extraction
- Forward and backward feature selection

Week 10: Model Evaluation and Hyperparameter Tuning

- Metrics: Accuracy, precision, recall (sensitivity), F1-score, specificity
- ROC (Receiver Operating Characteristic) curve and thresholding
- Cross-validation and grid search

Week 11: Unsupervised Learning

- Clustering techniques: k-means and hierarchical clustering
- Dendrograms and their interpretation

Week 12: Ensemble Models

- Random forest
- BAgging (Bootstrap Aggregating) and Boosting
- AdaBoost (Adaptive Boosting) and Gradient boosting

Week 13: Neural Networks and Project Presentations

- Fundamentals of Neural networks
- Project presentations

Week 14: Final Review and Conclusion

- Final exam preparation
- Project presentations
- Course wrap-up and next steps

Homeworks

Completing all assignments *independently* is essential for success in this class. You must strictly adhere to the homework submission requirements provided on iCollege.

Each submission should consist of the source code as Jupyter notebook file (with an extension ipynb). The submissions must be uploaded on iCollege dropbox into the relevant folder.

If the teaching assistant cannot open and read it, or if you submit a flawed document, the grade will be zero. It is your responsibility to ensure that each file opens correctly. Having submitted it on iCollege, go and try to open/download it. Make sure that you hit the "submit" button, and get a notification from iCollege that your submission has successfully gone through.

Your homework assignments will be checked by

Name	Email	Office hours
Abhiram Gaddam	agaddam2@student.gsu.edu	in Webex by appointment

Final Project

The final project is a team-based endeavor aimed at applying data science concepts to a real-world problem. Teams of 4–6 members will design and implement a data analytics solution over multiple phases, culminating in a presentation and final report. The project will count for **20%** of your final grade.

Project Phases

Phase 1 Proposal: Form teams and submit a project proposal, including the problem statement, dataset(s), and analytics goals.

Phase 2 Business Understanding: Define the business problem and finalize the dataset and analytics solution.

Phase 3 Data Preparation: Explore and preprocess the data. Select features.

Phase 4 Modeling and Evaluation: Test at least three categories of models and perform hyper-parameter optimization.

Phase 5 Communication: Present your findings in a 15-minute presentation and submit a final report.

Deliverables and Grading

- ✓ Processed Datasets (10 points): Submit raw and preprocessed datasets.
- ✓ Project Implementation (35 points): Provide Jupyter notebooks for data preprocessing and modeling.
- ✓ Presentation (30 points): Deliver a clear and professional group presentation, followed by a Q&A.
- ✓ Final Report (25 points): Submit a detailed PDF report explaining your process, findings, and recommendations.

Deadlines

- February 18$^{\text{th}}$: Proposal submission (via email to `kkuzmin1@gsu.edu` with CC to all members).
- April 16$^{\text{th}}$, April 21$^{\text{th}}$, April 23$^{\text{th}}$: Project presentations.
- April 25$^{\text{th}}$: Final report submission.

Further details and guidelines will be provided on iCollege.

Textbooks

Course materials should be sufficient to excel in this class. However, if you prefer additional resources, I recommend the following books:

1. Grokking Machine Learning by Luis G. Serrano (Manning Publications Co., 2021). ISBN: 978-1617295911
   A beginner-friendly introduction to machine learning concepts with clear explanations and visual examples.

2. Data Science from Scratch by Joel Grus (O'Reilly Media, Inc, 2019). ISBN: 978-1492041139
   Introduces data science fundamentals with Python, focusing on coding concepts from the ground up. While this book emphasizes implementation from scratch, in this class, we will primarily focus on using `scikit-learn` methods.

3. Fundamentals of Machine Learning for Predictive Data Analytics by John D. Kelleher, Brian Mac Namee, and Aoife D'Arcy (Second Edition, The MIT Press, 2015). ISBN: 978-0262029445
   A practical guide to machine learning techniques, with a focus on predictive analytics and practical applications.

4. An Introduction to Statistical Learning with Applications in Python by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor (Springer, 2023). ISBN: 978-3031387463
   An accessible introduction to statistical learning, covering Python-based implementations and real-world applications.

Books are listed in increasing order of rigor and difficulty.

Prerequisites

- You are expected to have taken 2510 Theoretical Foundations of Computer Science and 2720 Data Structures, and received a grade of C or higher in both classes.
- Basic knowledge of probability, algebra, and calculus is required.
- Programming in at least one programming language. Preferably in Python 3, since all examples in lectures and all homeworks will be given with expectation of some knowledge of Python 3.

Course Policies

Grading

The typical GSU grading scale will be used:

A+	A	A-	B+	B	B-	C+	C	C-	D	F
$[97, 100]$	$[93, 97)$	$[90, 93)$	$[87, 90)$	$[83, 87)$	$[80, 83)$	$[77, 80)$	$[73, 77)$	$[70, 73)$	$[60, 70)$	$[0, 60)$

Curving is left at the instructor's discretion and is (very) unlikely to happen.

The grade will count the assessments using the following proportions:

- **30%** of your grade will be based on homework assignments (5 total, with the lowest score dropped).
- **20%** of your grade will be based on the project (proposal due February 18th).
- **25%** of your grade will be based on the midterm, scheduled for regular class hours on Wednesday, February 26th (tentative).
- **25%** of your grade will be based on the final, held during exam week in the same classroom on Wednesday, April 30th, 4:15–6:00pm.

Makeup exams – midterm and final – are not given! Exceptions to this policy may be made only in case of a family or medical emergency, which must be clearly documented AND submitted through Professor Absence Notification system. Please note that to be eligible for makeup exams, your absence request *must be approved* by the Office of the Dean of Students.

Attendance

Attendance at the lectures is not mandatory but is strongly encouraged. This class is one of the most crucial for aspiring data scientists and data analysts. We will focus on discussing and solving numerous problems frequently encountered in technical job interviews.

Late Assignments

Late assignments may be accepted for no penalty if a valid excuse is communicated to the instructor before the deadline. If you exceed the deadline for more than 1 hour, your assignment will be penalized. The penalty is a 20% deduction for each extra 24 hours after the deadline. E.g., a homework scored 83% but submitted one day late (more than 1 hour and less than 24 hours late) will only receive 63% for the grade; the same homework submitted 2 days late (more than 24 hours and less than 48 hours late) will only receive 43%, and so on. Thus, even a perfect 100%-score submission which is more that four-day-late will receive no credit (but may receive some feedback).

Cheating

Do not cheat! While discussing problems with others is allowed, coding must be done independently. If you get stuck, you may research approaches online (e.g., stackoverflow.com) or use generative AI tools (e.g., ChatGPT). However, avoid blindly copying solutions – understand them first, then close the resource and implement the solution based on your understanding and memory.

Copying another student's work will result in severe penalties, including a grade of zero for the activity (homework/exam) for all students involved. In some cases of plagiarism, at the instructor's discretion, students may receive a failing grade for the course.

Students with Special Needs

Students requiring special accommodations are encouraged to schedule a meeting with the instructor as soon as possible. Prior to the meeting, please email your Faculty Notification Letter. Note that accommodations are not applied retroactively and will take effect only from the time they are granted.

Closing of the University

Sometimes the university closes because of weather conditions. To find out if the university is closed, go to gsu.edu or listen for an announcement on local TV or radio channels. If a class is canceled, planned activities will take place during the next class session.